

China Semiconductors

Montage Technology: MRDIMM deep dive



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Following our [Montage initiation](#) and [memory-interface chip industry primer](#), investors debate whether MRDIMM can evolve beyond a niche server-memory product and achieve meaningful penetration. This deep dive directly addresses that uncertainty through a bottom-up assessment of MRDIMM technology, workload suitability, platform readiness and adoption pathways. Our analysis suggests that MRDIMM solves a structural DDR bandwidth constraint emerging in the agentic AI era, while at the same time maintains high compatibility with existing DDR ecosystem and only introduce limited incremental cost. We therefore forecast MRDIMM penetration rising to 25%+ by 2030, **reinforcing our Outperform rating on Montage.**

MRDIMM offers the lowest-disruption path to addressing the CPU memory wall. By leveraging rank-level parallelism, MRDIMM doubles host-side bandwidth while preserving the existing DDR5 ecosystem. Unlike alternatives that require wider interfaces, additional channels or platform redesigns, MRDIMM concentrates complexity within module-level interface chips. Beyond throughput, higher bandwidth can materially reduce memory queuing delays under heavy load, improving system-level latency for memory-intensive workloads. Meanwhile, economics remain attractive: we estimate MRDIMM carries only a ~10% module premium versus RDIMM for a 128GB module, suggesting cost should not be a major adoption hurdle.

Agentic AI structurally increases the importance of DDR bandwidth and expands MRDIMM's addressable market. The industry is transitioning from compute scaling toward context scaling, where inference, retrieval, RAG, multi-step agent workflows, tool invocation and persistent memory increasingly depend on efficient context movement. In this architecture, DDR evolves from a passive capacity layer into an active bridge between HBM and storage-based context repositories. Meanwhile, CPU compute capability continues to scale faster than DDR bandwidth, compressing available bandwidth per unit of compute. Our analysis identifies AI host-node CPUs and agentic sandbox CPUs as structurally attractive MRDIMM deployment target, while recommendation, embedding, retrieval/RAG, CPU inference and data-movement-intensive workloads represent the most compelling use cases.

Our framework suggests MRDIMM can become a mainstream performance tier rather than a niche product. We believe adoption should be supported by rising deployment of AI host-node and sandbox CPUs, expanding memory-bound workloads, and increasing platform support from Intel, AMD and major DRAM suppliers. Under these assumptions, we forecast MRDIMM penetration could reach 25%+ by 2030, still conservative relative to historical DDR5 adoption trends.

We reiterate Outperform on Montage and emerge more constructive on the MRDIMM-driven upside opportunity. Our deep dive analysis strengthens our confidence that MRDIMM can achieve meaningful penetration over the remainder of the decade, supporting a memory-interface-chip TAM approaching USD 20bn by 2030. We therefore believe that the adoption of AMD Venice CPU in 2027 will start to attract more CSPs to move from RDIMM to MRDIMM, provide upside to MRDIMM penetration beyond 2027. 2026 will be a transition year when the CSP adoption likely will be mainly testing and thus MRDIMM penetration is expected to remain low.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	7 Sep 2026	Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
6809.HK (Montage)	O	HKD	279.00	520.00	NA	CNY	1.97	3.24	5.59	121.2	73.7	42.7
688008.CH (Montage)	O	CNY	197.82	400.00	51.9%	CNY	1.97	3.24	5.59	100.4	61.1	35.4
ASIAx			1,948.83									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate **Montage** Outperform, with A-share TP at CNY 400 and H-share TP at HKD 520.

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DETAILS

DECODING THE TECH BEHIND MRDIMM

FOUR FUNDAMENTAL PATHS TO BOOST MEMORY BANDWIDTH

The peak bandwidth between CPU and DRAM is measured by how many data can go through the CPU host, measure by **gigabyte per second (GB/s)**. If we break down the bandwidth calculation into different elements, it follows a simple equation:
CPU Bandwidth (GB/s) = native DRAM data rate (GT/s) x Rank count (#) x interface width (bits) x channel count (#) / 8

These four parameters capture different aspect of data transfer:

1. **Native DRAM data rate:** how fast data package moves out of DRAM. This is measured by giga transfers per second (**GT/s**). A transfer does not specify how many payload bits or bytes it carries, which will be defined by interface width.
2. **Rank count:** A **rank** is a group of DRAM devices selected together to supply data across the relevant interface. Think about this as the smallest package of data that can be transferred at a time coming out of the interface. RDIMM package can only run one rank at a time, while MRDIMM package can run two ranks at a time.
3. **Interface width** measures the number of payload bits transferred simultaneously. For server DRAM this is generally 64 bits. For HBM, the interface width will be much bigger, for example HBM3 has 1,024 bits, while HBM4 has 2,048 bits.
4. **Channel count** measures the number of independent memory paths whose bandwidth adds across the CPU.

Dividing by 8 converts bits into bytes, as 1 byte = 8 bits.

There four parts essentially capture how fast data package moves (1), how much data can be moved in one package (2,3), and how many packages can be delivered in parallel (4). Using cargo truck shipment in freight system as analogy Exhibit 1:

1. DRAM data rate is how fast a truck can move;
2. Rank count is how many stacks of cargo container per truck (package delivery in parallel);
3. Interface width is how long the cargo container is (smallest package);
4. Channel count is how many independent routes trucks can move in parallel.

AMD's EPYC 9006 (Venice) server CPU provides an illustrative example. Its top memory configuration supports 16 DDR5 channels and Multiplexed Rank Dual Inline Memory Module (MRDIMM) modules at 12,800 MT/s (6,400 MT/s with 2 ranks in parallel), equivalent to 12.8 GT/s. Each channel carries 64 bits per offloading operation. It therefore reaches 1.6TB/s of peak bandwidth per CPU, with the calculation process below:

Venice peak BW = 6,400 MT/s x 2 ranks x 64 bits x 16 channels / 8 = 1.6 TB/s

For MRDIMM (Multiplexed Rank Dual In-line Memory Module), effective data rate measures transfers per second on each data pin. The effective data rate can be decomposed further to two sub factors:

Effective host data rate = native DRAM data rate x rank-stream aggregation factor.

Conventional RDIMMs transfer data from one rank at a time. MRDIMM combines two concurrent rank streams through its data buffers, allowing the host interface to operate at approximately twice the data rate of either rank. Two rank-side streams at 6,400 MT/s can therefore feed a host interface at 12,800 MT/s, which is the key technology innovation to deliver higher bandwidth. As a result, MRDIMM use the same memory die (the DRAM data rate is the same as RDIMM), but delivers 2X effective host data rate.

EXHIBIT 1: The freight system offers the simplest analogy. Higher bandwidth equals to more cargo received per second. Fundamentally, there are four routes to deliver more cargo

$$\text{BW (GB/s)} = \text{native DRAM data rate (GT/s)} \times \text{Rank count(\#)} \times \text{Interface width (bits)} \times \text{Channel count} \div 8$$

FREIGHT LEVER

1 Increase truck speed

Same truck and container size

2 Stack containers vertically

Two standard payloads per truck

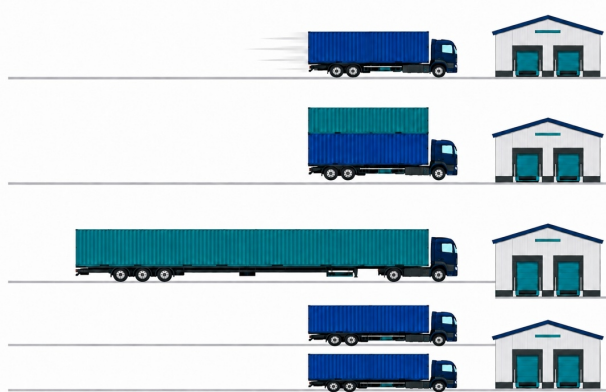
3 Lengthen truck + container

One larger payload per departure

4 Add independent routes

More roads operating in parallel

CARGO IN TRANSIT → RECEIVING END (CPU)



MEMORY BW LEVER

DDR generation migration

Higher native DRAM data rate, to fulfill more transfers per second

MRDIMM

Rank aggregation raises host rate

HBM

Wider aggregate data interface

More DDR channels per socket

Through a CPU upgrade, more independent data paths

conceptual analogy; MRDIMM serializes two rank streams onto a faster host link of unchanged width, and it does not widen the interface or physically stack DRAM
Source: Bernstein analysis

As a result, there are four distinct hardware routes to expand memory bandwidth: increase native DDR data rate, increase rank-stream aggregation, widen the interface, or add more channels. MRDIMM doubles the host interface speed by exploiting rank-level parallelism, offering the lowest-disruption and cost-efficient path

Route 1: increase native DRAM data rate; deliver trucks more frequently. Faster DRAM generations shorten the interval between transfers and raise the data rate of the DDR cells. This route depends on DRAM process upgrade, I/O signaling and signal integrity advances. The scaling cadence is therefore tied to DRAM generations, while power and timing margins become

progressively harder at higher speeds. DDR5 speed improvement is approaching the ceiling.

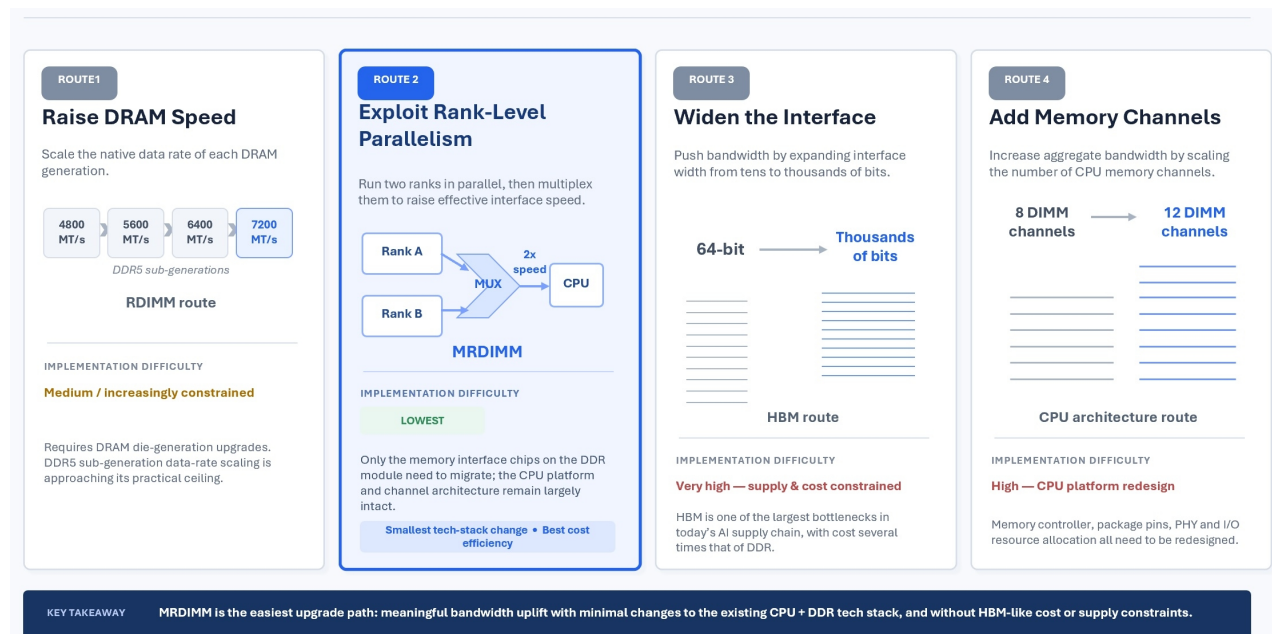
Route 2: exploit rank-level parallelism; stack two containers vertically on each truck. Two depots supply separate containers concurrently, and MRDIMM stack these two containers vertically on one truck. Each container retains its original size. The doubled cargo is converted into a faster sequence of transfers, increasing effective host data rate. The MRCD coordinates rank access, and MDBs multiplex the two rank-side streams onto the faster host interface. The channel count and 64-bit width remain unchanged, and DRAM cells do not need to run at the 2x host rate. MRDIMM instead double the trucks traffic in one time slot on the existing roads, resulting in higher loaded throughput.

Route 3: widen the interface; lengthen the truck and its container. Each delivery carries one larger payload, which the warehouse accepts in a single unloading operation through an expanded receiving interface. Unlike Route 2's two vertically-stacked containers, this Route operates one longer container to transfer more bits simultaneously. It expands interface width, while allowing transfer frequency to remain unchanged. Enabled by dense I/O connections and close physical placement, HBM illustrates this approach through thousands of bits wide versus 64 bits for a DDR channel. The result is very high bandwidth, but it requires close memory placement, advanced packaging and a different memory supply and cost structure. Capacity and serviceability are also trade-offs.

Route 4: add memory channels; open more delivery lanes. Each lane carries an independent stream of trucks, so adding lanes increases aggregate cargo received. However, every channel requires another memory-controller interface, PHY resources, CPU package pins, socket connectivity, and motherboard redesign. The bandwidth uplift is direct, but the change sits at the CPU-platform level and consumes scarce package and board area.

The location of the architectural change determines disruption. Route 1 waits for the next generation of DRAM process node. Route 3 redesigns the CPU package, socket and board. Route 4 changes the physical memory architecture and packaging ecosystem. MRDIMM takes the Route 2 and concentrates most incremental complexity on interface chips on the DIMM module. The CPU memory controllers must still support the MRDIMM protocol and the system must qualify higher module power and thermals, but channel count, bus width and the broader DDR5 module ecosystem remain familiar. Hence, **MRDIMM is the narrowest architectural intervention that can deliver a step-change in host-side bandwidth.**

EXHIBIT 2: Among four tech routes to increase memory bandwidth, MRDIMM doubles the host interface speed by exploiting rank-level parallelism, offering the lowest-disruption and cost-efficient path

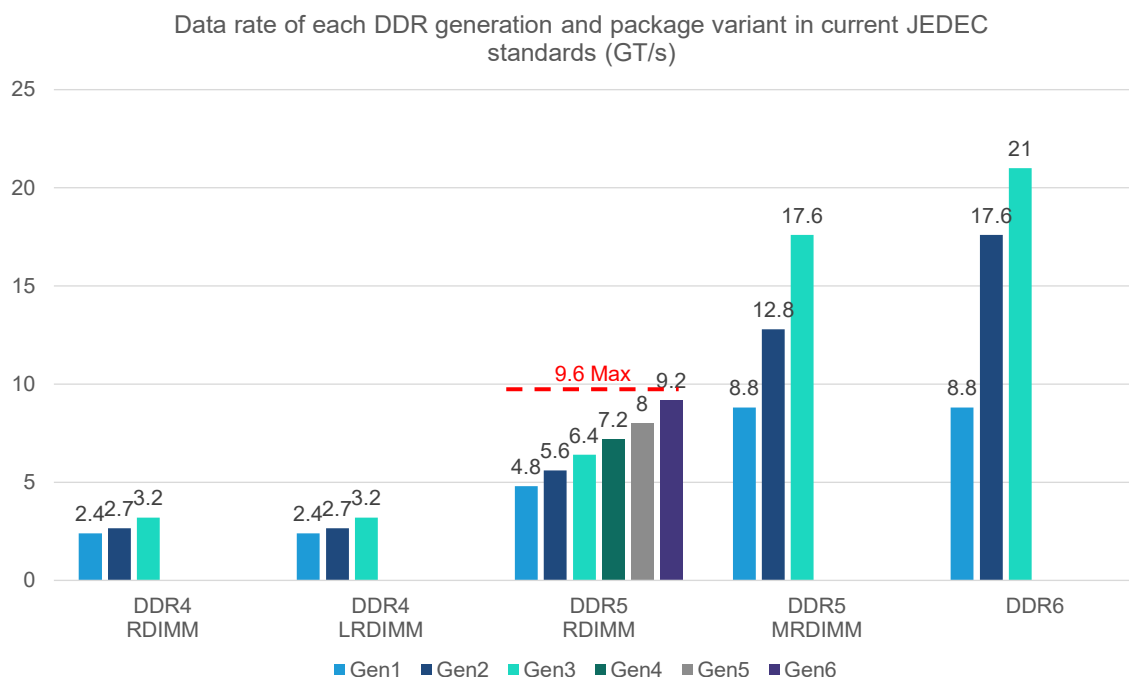


Source: company reports, Bernstein analysis

MRDIMM DOUBLES HOST-INTERFACE SPEED BY EXPLOITING RANK PARALLELISM

In JEDEC's current roadmap, DDR5 RDIMM progresses from 4,800 MT/s in Gen1 to 9,200 MT/s in Gen6, while MRDIMM standards advance from 8,800 MT/s to 17,600 MT/s across three generations. MRDIMM therefore reaches DDR6-class data rates, while continues to use DDR5 DRAM technology and a DIMM-based physical architecture.

EXHIBIT 3: DDR5 MRDIMM doubles RDIMM bandwidth and achieves DDR6-class data rates, while leveraging the existing DDR5 die and physical architecture

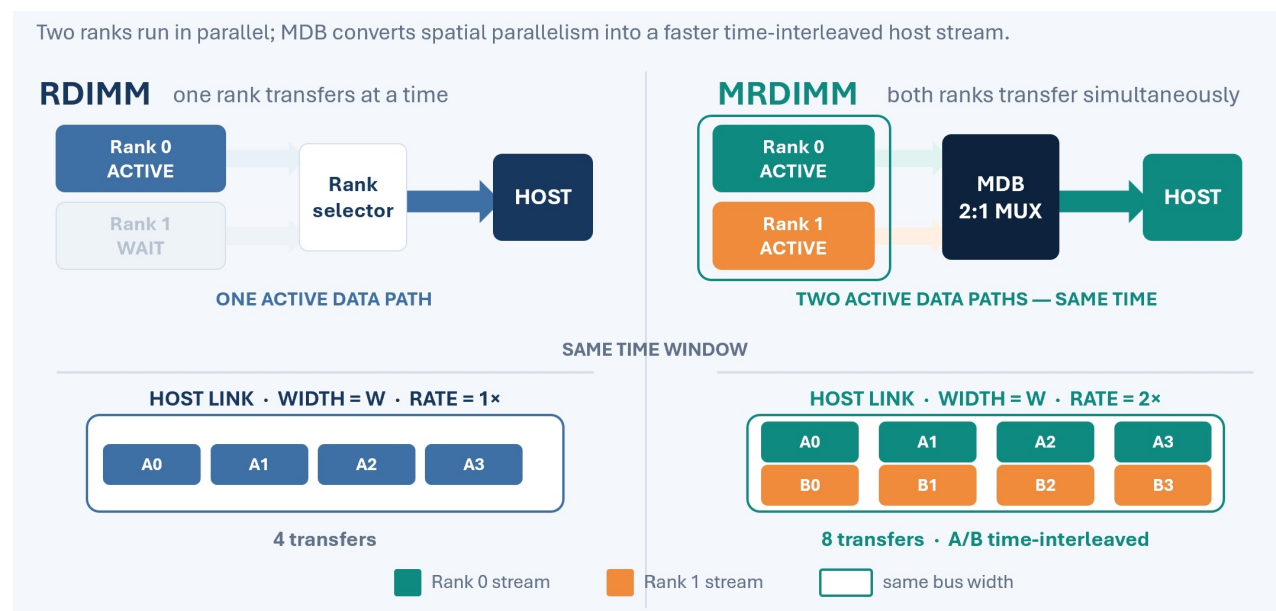


JEDEC's earlier long-term roadmap targeted up to 17.6GT/s for MRDIMM Gen3, while current Gen3 memory interface chipset by Renesas and Montage support 16GT/s

Source: JEDEC, Bernstein analysis

The mechanism for MRDIMM is rank-level parallelism. A conventional RDIMM selects one rank at a time, leaving the second rank idle during the same transfer window. MRDIMM's MRCD can address two ranks concurrently, while the MDBs multiplex and time-interleave the two data streams before transmitting them to the host. The channel remains 64 bits wide, but it carries twice as many transfers in the same period.

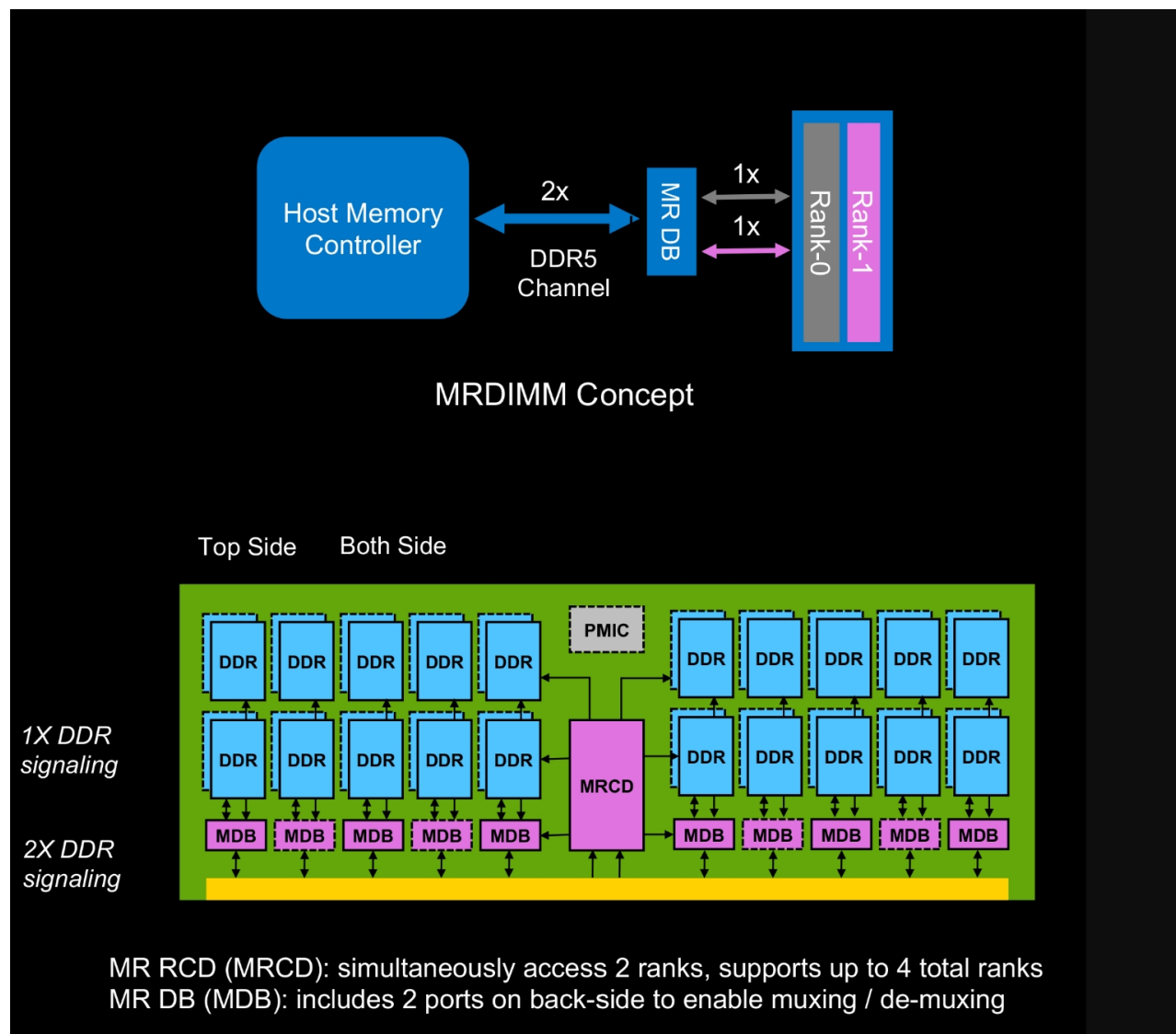
EXHIBIT 4: **MRCD allows simultaneous access to two DDR ranks, and MDB performs muxing and demuxing. Together, two DDR ranks run in a multiplexed mode to double the speed and correspondingly the bandwidth, without broadening the host interface channel width (still in 64bit)**



Source: JEDEC, Bernstein analysis

The module-level implementation explains the silicon-content uplift. One MRCD controls command/ address/ clock signals and coordinates rank access. Ten units of MDBs sit closer to the DRAM dies, each with two rank-side ports and one faster host-side path for multiplexing and de-multiplexing. The design supports simultaneous access to two ranks and up to four total ranks.

EXHIBIT 5: A set of “1 MRCD + 10 MDB” chips allow two DDR ranks to operate in parallel and doubles host-side transfer rate



Source: Micron, Bernstein analysis

MRDIMM OPTIMIZES TASK-LEVEL LATENCY BY REDUCING THE QUEUING DELAY

Although MRDIMM introduces a small amount of additional intrinsic latency due to the active interface chips in its data path, its higher interface bandwidth can significantly reduce task-level latency that is often dominated by queuing effects. Under light traffic, RDIMMs may retain a modest latency advantage because MRDIMMs add multiplexing, buffering, and retiming overhead. However, the benefits of MRDIMM become increasingly apparent as memory utilization rises. Near saturation, queuing delay in RDIMMs grows nonlinearly, whereas MRDIMM's higher service bandwidth helps maintain lower utilization and shorter memory-request queues.

High-concurrency workloads, such as recommendation systems, retrieval applications, CPU inference, and data-movement-intensive tasks, can benefit substantially because both aggregate throughput and tail latency are strongly influenced by queue depth. Consequently, MRDIMM's primary value proposition is the ability to deliver higher throughput while maintaining lower latency under high utilization.

EXHIBIT 6: With sufficient bandwidth, MRDIMM can substantially reduce queuing delay, resulting in system-level latency improvement for high-load tasks

MRDIMM shifts the memory queuing wall to the right

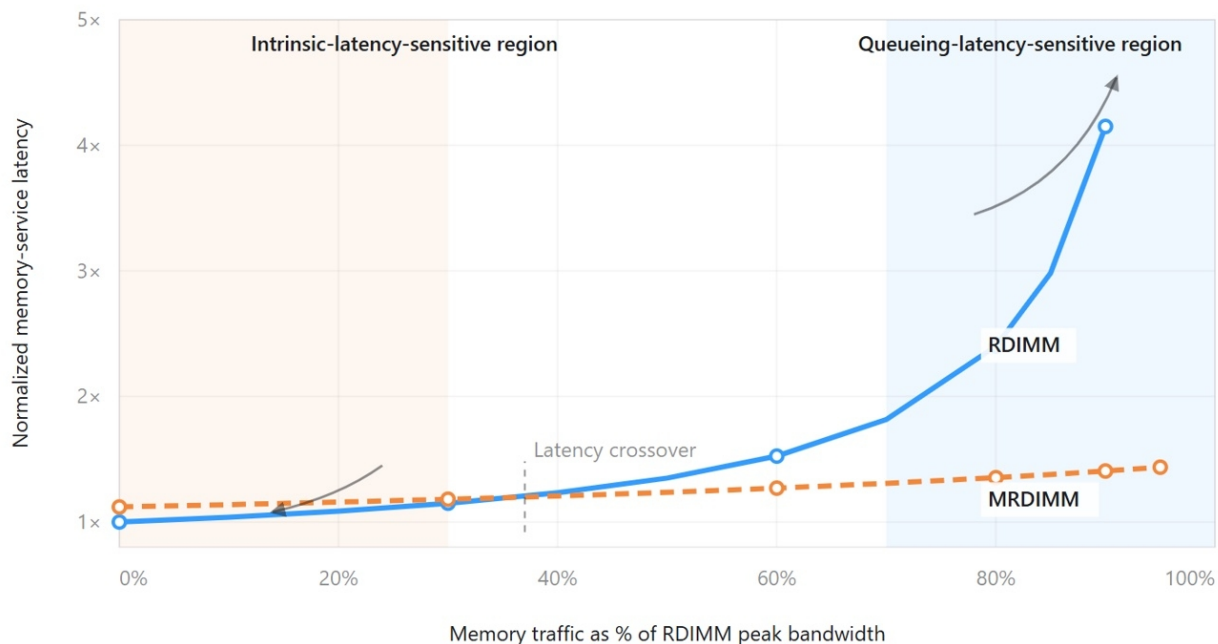
Higher service bandwidth trades modest unloaded latency for materially lower loaded latency.

● Low load: intrinsic latency matters

With little queueing, RDIMM retains a small latency advantage; MRDIMM carries added mux and buffer-path overhead.

● High load: queueing latency dominates

RDIMM latency rises non-linearly near saturation, while MRDIMM's higher bandwidth keeps utilization—and queueing—lower.



Illustrative normalized model, not measured benchmark data. MRDIMM effective service bandwidth assumed at ~2× RDIMM.

The line chart is a illustrative normalized model, not measured benchmark data. MRDIMM effective bandwidth is assumed at 2x RDIMM
Source: company reports, Bernstein analysis

POWER, THERMAL DENSITY, AND PHYSICAL SPACE ARE THE PRIMARY TRADE-OFFS

MRDIMM buys bandwidth through active electronics and higher I/O activity. The MRCD, ten MDBs, and simultaneous rank operation **increase module power and thermal density versus RDIMM, making cooling design the principal architectural constraint**. Tall-form-factor (TFF) MRDIMM expands PCB and cooling surface area to relieve local thermal challenges, but consumes more system space and can complicate airflow or liquid-cooling layouts.

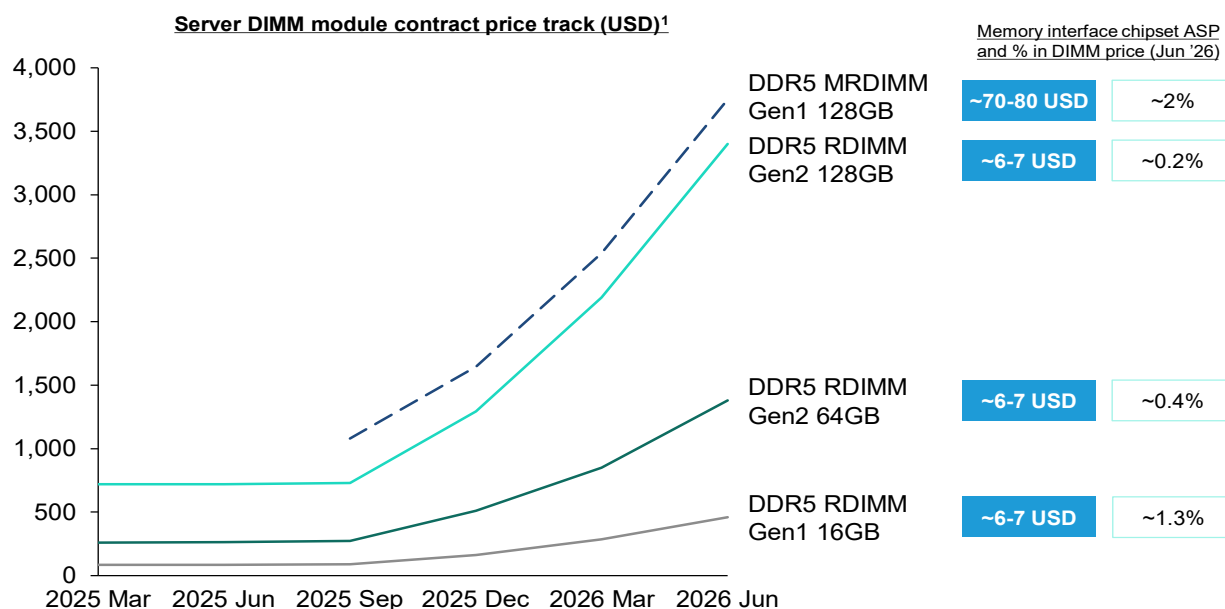
However, module watts alone do not determine the system-level efficiency outcome. If higher DDR bandwidth raises CPU utilization, reduces request queuing or protects the utilization of much more expensive accelerators, the system can still deliver attractive workload throughput per watt. Nevertheless, we do not assume this benefit for every workload; power-constrained and low-bandwidth systems can rationally remain on RDIMM or select a more compact architecture.

MRDIMM REQUIRES MODEST INCREMENTAL MODULE BOM

The additional BOM derived from interface chips is only LSD in the DRAM module ASP. The MRDIMM interface chipset sold in USD70-80 represents only 2% in the module price of a 128GB DDR5 MRDIMM at present. The extra silicon is substantial for interface-chip suppliers, but remains limited for the server buyers.

The percentage premium also falls mechanically, when DRAM prices hike. Because the interface-cost increment is relatively fixed, while DRAM dies dominate total module value. We estimate MRDIMM currently carries a USD300-350 premium over RDIMM at the same capacity, equivalent to roughly 10% at current 128GB module price. This narrows the adoption barrier from economic perspective, as the bandwidth problem becomes more acute.

EXHIBIT 7: Memory interface chips account for only an LSD portion of module costs. Rising memory prices are also reducing MRDIMM's premium over RDIMM to around 10%, reducing adoption barriers of MRDIMM



MRDIMM remains a niche product without an established contract pricing track record. The MRDIMM pricing shown in the chart is an estimate based on public information; in general, we estimate MRDIMM enjoys a \$300-350 markup over RDIMM at the same capacity

Source: TrendForce, Bernstein analysis and estimates

WHY DDR BANDWIDTH DEMAND STRUCTURALLY RISES IN THE AGENTIC AI ERA

FROM COMPUTING SCALING TO CONTEXT SCALING: IMPORTANCE OF DDR ELEVATED IN THE MEMORY TIERING

The industry's focus in AI infrastructure is gradually shifting from pure compute scaling toward context delivery.

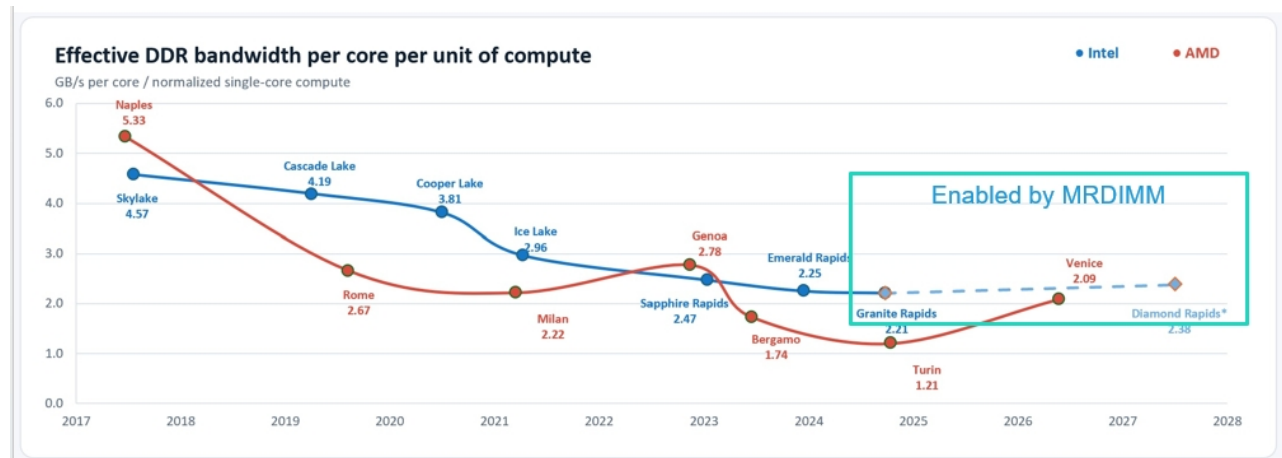
In traditional AI systems, memory was primarily used to store model weights and feed GPUs with sufficient data to maximize compute utilization. However, the rise of inference-centric and agentic AI workloads is creating a new bottleneck: the ability to efficiently store, retrieve, and manage large amounts of contextual information. Long-context reasoning, multi-step agent workflows, RAG retrieval, external tool invocation and persistent user memory all require AI systems to continuously access and update context beyond model parameters, making memory and storage performance increasingly critical to overall system efficiency.

As a result, AI infrastructure is evolving toward a more sophisticated memory hierarchy spanning HBM, DDR and storage-based context memory layers. And DDR plays a growing role as the bridge between HBM and storage-based context repositories. Rather than serving merely as a capacity expansion tier, DDR is becoming an active component of the context delivery pipeline, storing prompt histories, KV caches, retrieved knowledge and agent working memory. We believe this trend extends the role of DDR beyond serving traditional general-purpose servers and increases the strategic importance of DDR in AI systems. In this architecture, improving memory bandwidth and capacity is no longer just about supporting model execution, but increasingly about enabling low-latency access to context at scale, which is essential for sustaining high GPU utilization and efficient agentic reasoning.

The CPU memory wall reflects the widening gap between aggregate compute throughput and memory bandwidth.

Core counts per CPU continue to rise and per-core compute capability also improves, so each GB/s of DDR bandwidth has to support materially more total compute across successive CPU generations. Socket-level bandwidth increases as CPUs add channels and adopt faster DDR, but it does not scale at the same rate as the compound effect of core-count and per-core-performance uplift.

EXHIBIT 8: Effective DDR bandwidth for per unit of compute compresses across CPU generations, since the CPU core count and single-core compute scale faster than socket-level DDR bandwidth



Source: AMD and Intel companies reports, Bernstein analysis and estimates

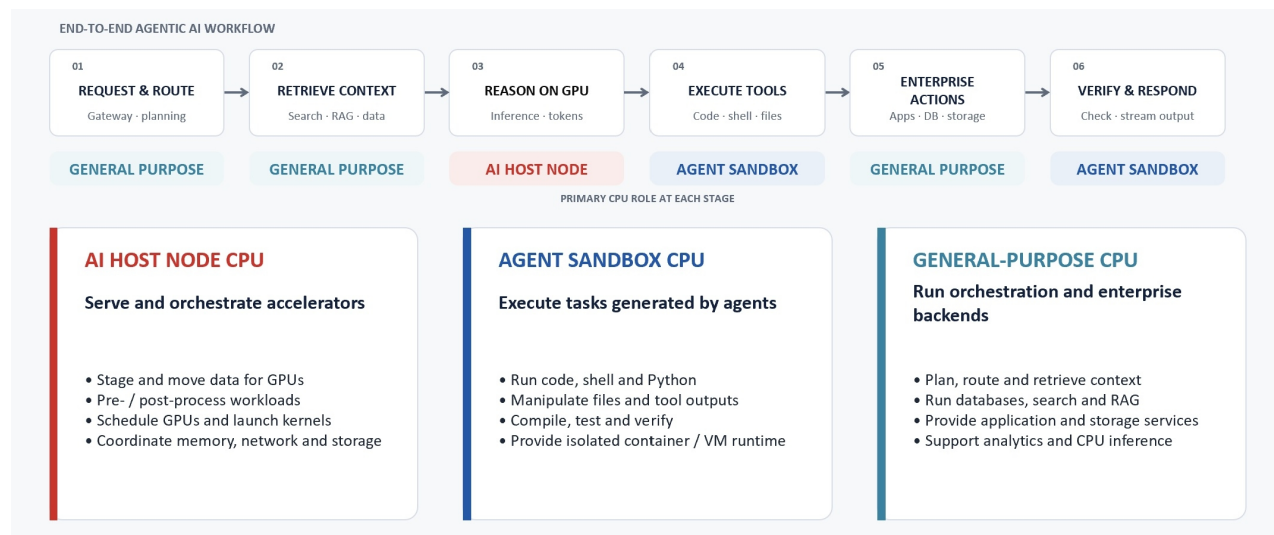
This compression does not make every compute workload memory-bound. It raises the probability that workloads with large active data sets, low arithmetic intensity, irregular access and high concurrency encounter DDR contention. MRDIMM becomes the most valuable, when the application already has sufficient CPU compute and I/O resources but cannot convert them into throughput because the memory subsystem saturates first.

AGENTIC AI ENGAGES THREE CPU ROLES WITH DIFFERENT DDR EXPOSURE

Agentic AI engages three types of CPU roles across the end-to-end workflow: **host-node** CPUs serve and orchestrate accelerators; **sandbox** CPUs execute tasks generated by agents; and **general-purpose** CPUs run retrieval, databases, enterprise applications, storage and CPU inference. CPU role determines what the processor does, while workload

characteristics determine whether DDR bandwidth becomes the bottleneck.

EXHIBIT 9: AMD classifies three types of CPU engaged in the agentic AI workflow. Their sensitivity levels to DDR bandwidth are different



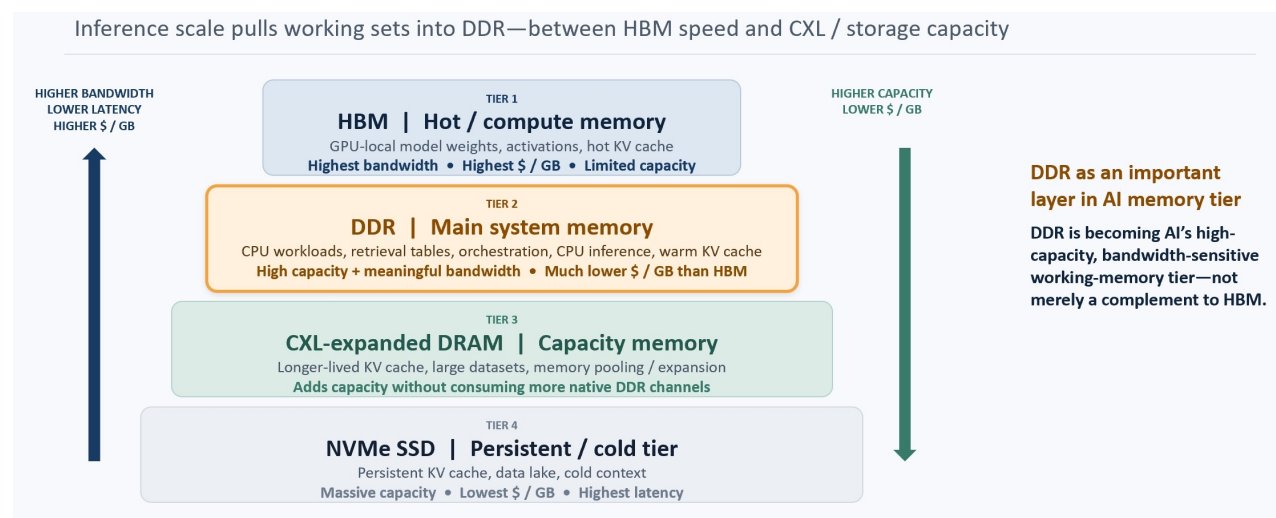
Source: AMD reports, Bernstein analysis

Host-node CPUs have the most structural MRDIMM fit

Host nodes sit on the system data path that stages and move data, launches accelerator kernels, and coordinates memory, networking and storage. DDR bandwidth can therefore become a system-level constraint: if the host node cannot prepare or move data fast enough, expensive GPUs wait. The economics value of relieving the bottleneck can exceed the direct CPU-performance benefit, because buyers prioritize accelerators utilization.

AI memory is also evolving from an HBM-centric view into a hierarchy. HBM remains the hot, compute-local tier; DDR provides high-capacity main-system memory and can hold retrieval tables, orchestration state, CPU-side inference data, intermediate buffers and warm working sets; CXL-expanded DRAM adds capacity; and NVMe remains the persistent cold tier. DDR complements HBM, and its relevance rise as context length, concurrency, and model dimensions expand the aggregate working set.

EXHIBIT 10: AI memory is evolving from an HBM-centric architecture into a multi-tier hierarchy, where DDR is becoming increasingly relevant for inference tasks



Source: Micron and SK hynix reports, Bernstein analysis

In a conventional PCIe-based host architecture, the CPU-GPU interconnect often becomes the bottleneck before DDR bandwidth is fully utilized, limiting the incremental benefit of faster memory. However, as next-generation CPU-GPU interconnects such as NVLink-C2C and UALink (UB) emerge and provide substantially higher bandwidth, the bottleneck is increasingly likely to shift to the DDR subsystem. Meanwhile, the opportunity cost of GPU underutilization continues to rise as GPU compute performance and pricing increase. This strengthens the incentive to provision sufficient memory bandwidth and avoid GPU stalls caused by data starvation. Consequently, we view host nodes as the most structurally attractive deployment opportunity for MRDIMM.

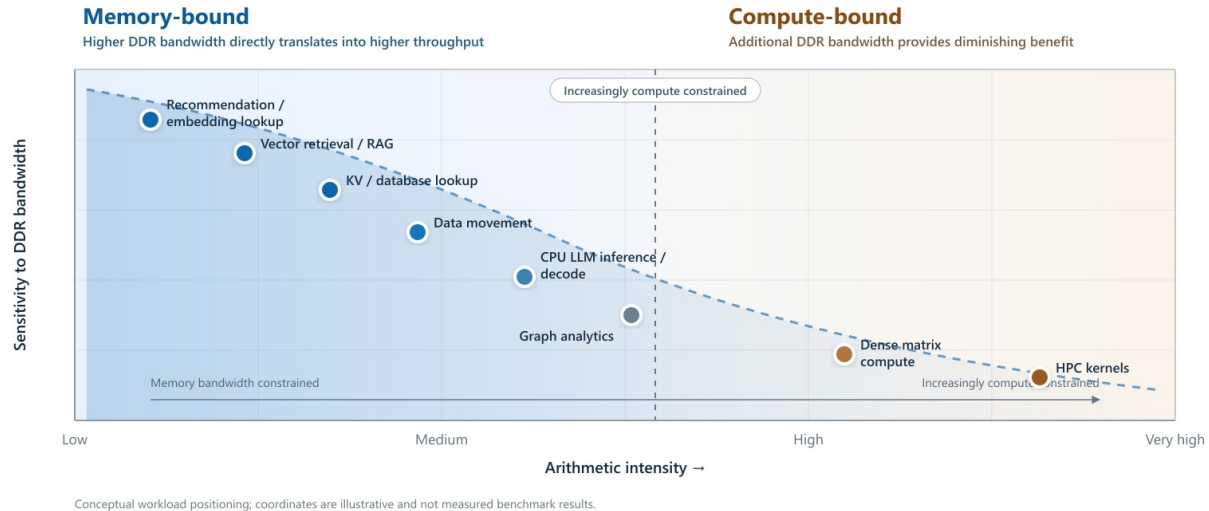
Sandbox and General-purpose CPUs adoption of MRDIMM depends on workload mix

Sandbox CPUs execute heterogeneous tasks, including coding, compilation, file manipulation, browser and API calls, retrieval, analytics, and CPU inference. Many tasks are compute-, network-, storage- or latency-bound and receive limited benefit from additional DDR bandwidth. Others, particularly vector retrieval, embedding lookup, large data transforms, graph analytics, CPU LLM inference and memory movement, can saturate DDR channels. General-purpose CPUs show the same dispersion, because their workload universe spans web and application servers as well as recommendation, databases, search and in-memory analytics.

EXHIBIT 11: Sandbox and GP CPUs execute highly heterogeneous workloads. MRDIMM creates the most value in low-arithmetic-intensity workloads where data movement, not computation, constrains throughput

MRDIMM creates the most value for memory-bound workloads

Workloads with low arithmetic intensity move substantially more data per unit of computation, making system throughput more sensitive to DDR bandwidth. As arithmetic intensity rises, additional DDR bandwidth delivers diminishing performance benefit.

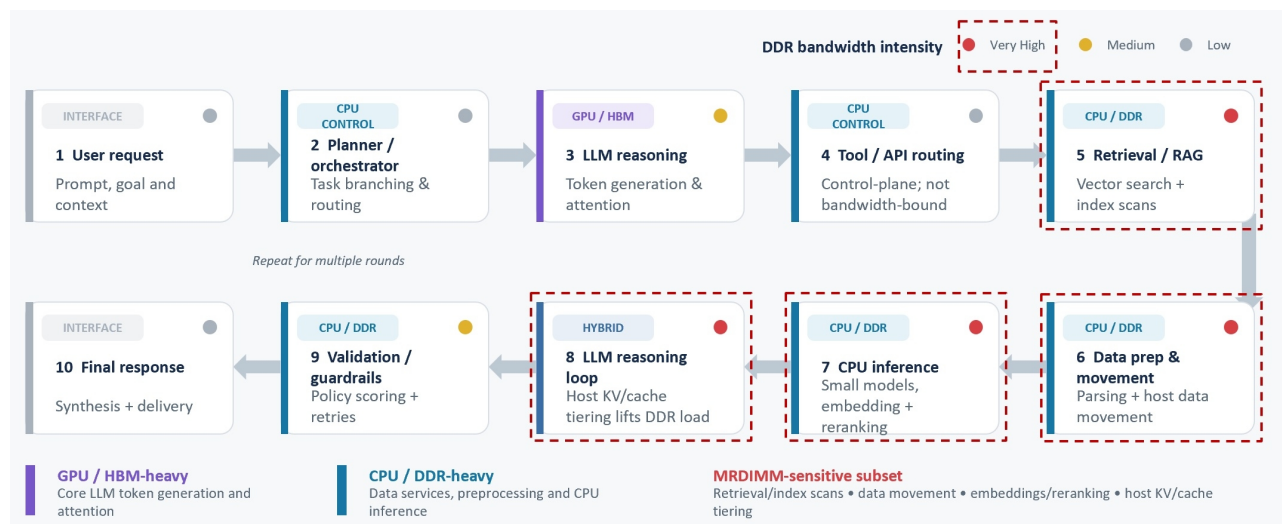


MRDIMM creates the most value where CPUs spend more time moving data than computing on it.

Source: company reports, Bernstein analysis

Within an agentic AI workflow, DDR bandwidth requirements can vary significantly across stages. Functions such as user routing and GPU-resident model inference typically exhibit limited direct sensitivity to DDR bandwidth, whereas **retrieval/RAG, data preparation and movement, CPU-based LLM inference, host-side KV cache tiering, and response validation can generate substantial memory traffic.** As a result, a single agentic workflow may encompass both MRDIMM-suitable and MRDIMM-insensitive stages. This suggests that MRDIMM deployment is more likely to be selective, with partial attachment to bandwidth-intensive nodes, rather than universal adoption across the entire system.

EXHIBIT 12: Several stages in the Agentic AI Workflow are DDR-BW-sensitive, creating ideal MRDIMM use cases within a heterogeneous task chain

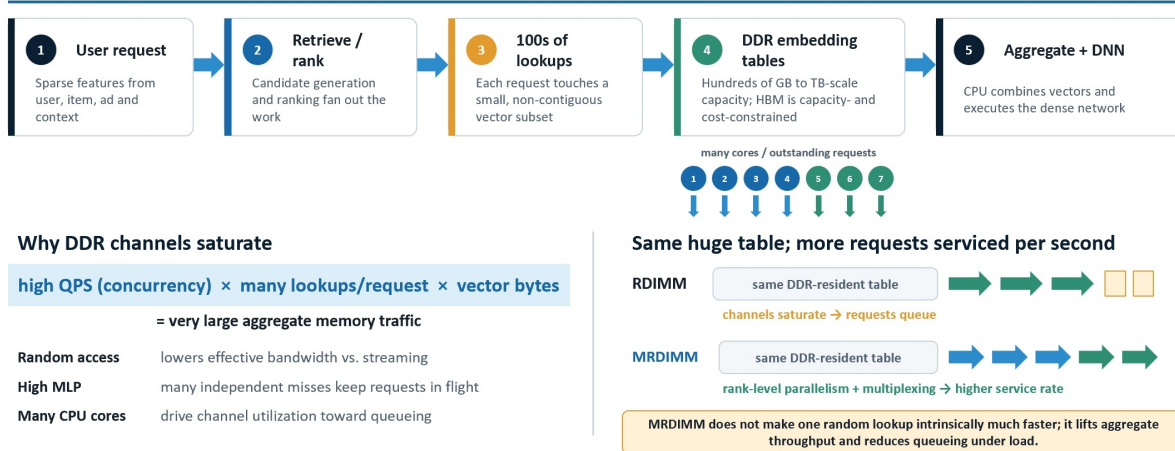


Source: company reports, Bernstein analysis

Recommendation and embedding represent the most ideal application. Their tables can exceed cache capacity by orders of magnitude, queries touch many sparse vectors with limited locality, and high QPS creates large numbers of independent memory requests. The bottleneck is aggregate random-access traffic under concurrency, not the bytes fetched by one request. MRDIMM does not make one random lookup intrinsically faster, but it can service more lookups in parallel and prevent the request queue from exploding under load.

EXHIBIT 13: Massive recommendation and embedding workloads combine random access, high concurrency, and DDR-restricted capacity, making them attractive MRDIMM workload profile

The bottleneck is aggregate random-access traffic under concurrency—not the bytes fetched by one query



INVESTMENT TAKEAWAY

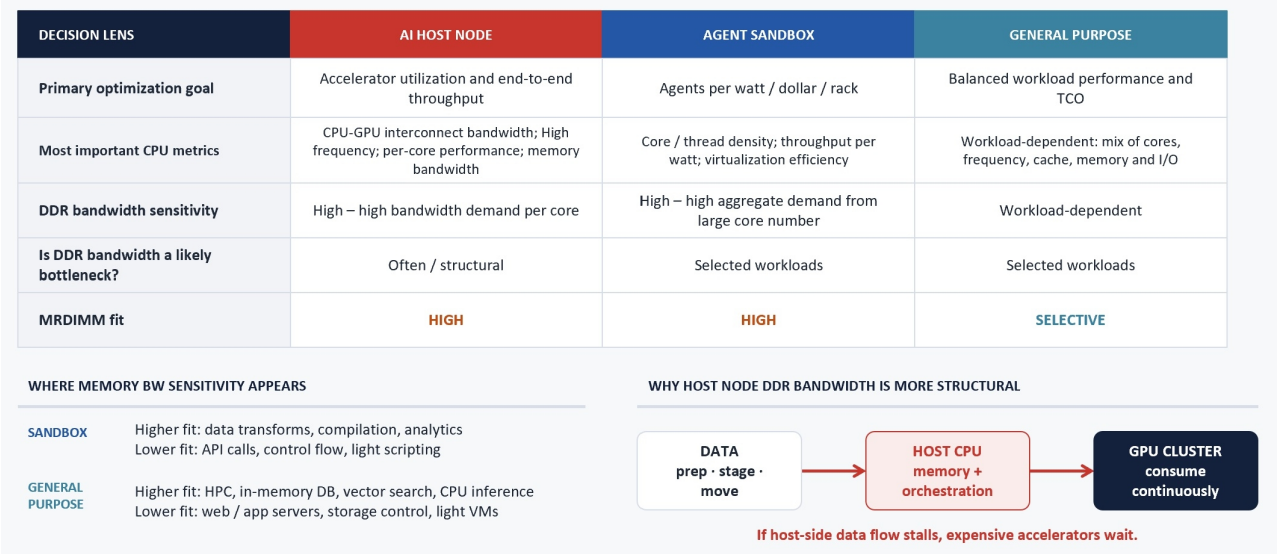
The MRDIMM fit is driven by high concurrency × many independent random lookups × massive DDR-resident capacity.

Source: Intel, Meta, Bernstein analysis

WE FORECAST MRDIMM SOCKET PENETRATION TO RISE TO 25%+ BY 2030

Host node CPUs represent the most structurally consistent MRDIMM opportunity, while sandbox and general-purpose CPUs broaden the addressable market in certain workloads where DDR bandwidth is the bottleneck to constrain throughput. Hence, we assume different MRDIMM attach rates across three CPU roles.

EXHIBIT 14: **Host node CPUs are structurally the most DDR-BW-sensitive; sandbox and general-purpose CPUs are workload-dependent, with recommendation, embedding, retrieval/RAG, and CPU LLM inference particularly attractive MRDIMM use cases**



Source: AMD reports, Bernstein analysis

We model overall MRDIMM socket penetration using two key variables for each CPU role: the global server CPU socket mix and the MRDIMM adoption rate within that specific CPU category. Role-level adoption is determined by several factors, including the share of DDR bandwidth-sensitive workloads, platform and memory-controller support, and customers' budget sensitivity.

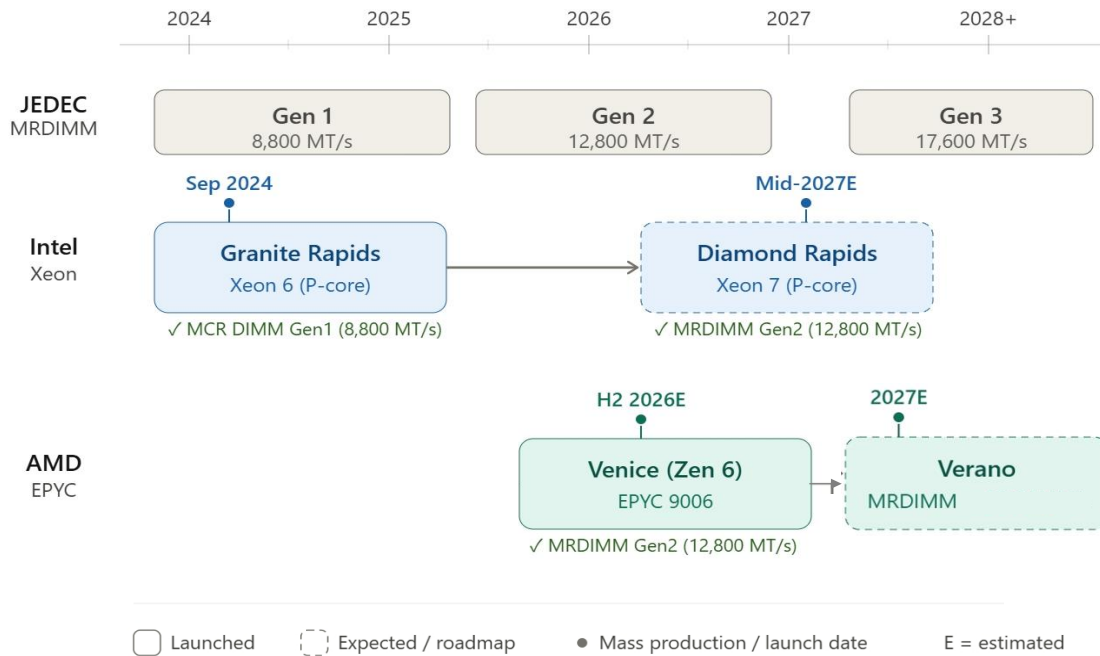
We expect penetration growth to be driven primarily by the increasing share of host-node and sandbox CPUs within the server installed base. While MRDIMM adoption should increase across all three CPU categories, the pace of penetration is likely to vary meaningfully by role. We anticipate the strongest and most structural adoption in host nodes, with more selective deployment in sandbox and general-purpose CPUs. Even under these relatively conservative assumptions, **we forecast overall MRDIMM penetration to increase roughly tenfold, from 2-3% in 2026 to 25%+ by 2030.**

CURRENT PROGRESS OF MRDIMM SUPPORT AND ADOPTION

CPU platform and DRAM roadmaps broaden the addressable pool of MRDIMM, while CSP workload mix determines its attach rate.

CPU support is a prerequisite. Intel initiated the migration since Grantie Rapids, supporting Gen1 MCR DIMM at 8,800 MT/s in Sep 2024. Diamond Rapids plans to support Gen2 MRDIMM in mid-2027. AMD enters directly at Gen2 with the latest-introduced Venice and plans to extend support in Verano in 2027E. This materially expands the set of high-end x86 sockets that can qualify MRDIMM.

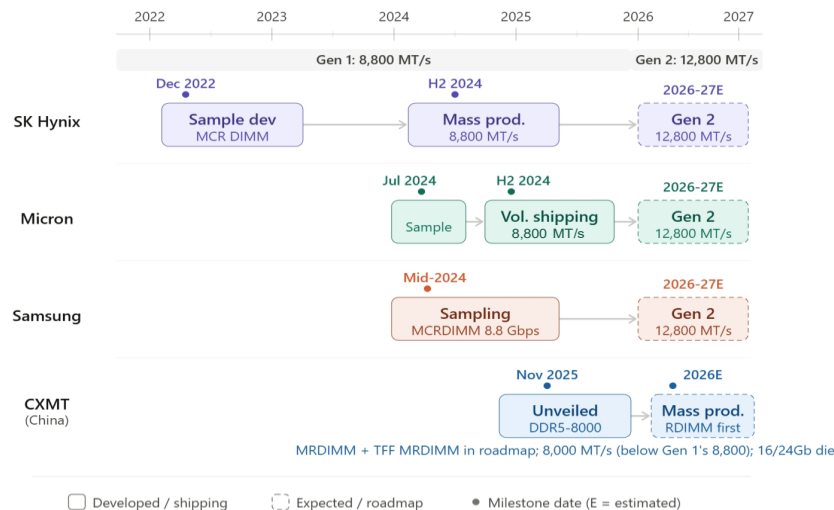
EXHIBIT 15: Intel led the migration to MRDIMM with Granite Rapids in 2H24, while AMD enters directly to MRDIMM Gen2 in 2026, with Venice as the first MRDIMM-capable platform



Source: Intel and AMD reports, Bernstein analysis

The memory supply chain is developing in parallel. Global leading vendors all have entered mass production stage. Meanwhile, CXMT also has introduced server-level DDR5 products and unveiled mass production plan for RDIMM as the first step. Multi-vendor MRDIMM support reduces supply concentration and gives CPU, OEM, and CSP customers greater confidence to invest in qualification.

EXHIBIT 16: DRAM suppliers have established solid MRDIMM product roadmaps; notably, CXMT is rapidly catching up with global leaders



Source: SK Hynix, Samsung, Micron, CXMT reports, Bernstein analysis

CSP potential adoption has showed early signals. Public deployment of Xeon 6 or Venice broadens the potential MRDIMM pool, though it does not prove that CSP customers deploy MRDIMM in every Xeon 6 or Venice supported server. Azure provides the most visible confirmation across both compatible CPU roadmaps, while AWS, Google Cloud, and Oracle have confirmed Xeon 6 adoption and memory-heavy workloads that support a high readiness assessment. Meta and ByteDance operate massive workloads of recommendation, ranking, and retrieval, theoretically having the highest incentive to adopt MRDIMM.

EXHIBIT 17: The adoption of Xeon 6 and AMD Venice by CSPs is a signal for the potential MRDIMM deployment, while workload mix determines actual attach rate. Azure is currently the most visible confirmed adopter of both MRDIMM-compatible CPU platforms

	CSP	Xeon 6 adoption	AMD Venice adoption	Memory-bound workload	MRDIMM current adoption probability
1	AWS	Confirmed – R8i / R8i-flex on custom Intel Xeon 6; X8i GA (Jan-26), marketed for very high memory bandwidth	Not publicly confirmed	Memory-optimized DB, in-memory cache, AI inference	High
2	Google Cloud	Confirmed – C4 GA on Xeon 6; X5 announced for large-memory SAP/HANA-style workloads	Not publicly confirmed	ML recommendation, large-memory DB / analytics	High
3	Microsoft Azure	Confirmed – Didsv7 / related Granite Rapids VMs	Confirmed – HDv2 and HXv2 on 6th Gen AMD EPYC ‘Venice’	AI / data processing, HPC, EDA, memory-sensitive VMs	Very High
4	Oracle OCI	Confirmed – X12 Standard Acceleron on Intel Xeon 6	Not publicly confirmed	DB, HPC, CPU inference, agentic AI	High
5	Meta	No public confirmation	No public confirmation	Recommendation / ranking, large-scale data serving	Medium-High
6	Alibaba Cloud	Platform readiness / no instance confirmation (Alibaba Cloud Linux 3 supports Xeon 6); no public instance confirmation	Not publicly confirmed	Cloud DB, analytics, AI services	Medium
7	Tencent Cloud	No public confirmation	Not publicly confirmed	Cloud / recommendation / data services	Medium
8	ByteDance	No public cloud-instance confirmation	Not publicly confirmed	Recommendation, embedding lookup, retrieval	Medium-High
9	Baidu AI Cloud	No public confirmation: public cloud materials still mostly reference earlier Xeon generations	Not publicly confirmed	Search / AI / retrieval / DB	Medium

MRDIMM readiness is based on our assessment

Source: companies reports, Bernstein analysis

MRDIMM VS SOCAMM2

TECH COMPARISON OF TWO DDR PACKAGE FORMATS FOR SERVERS

MRDIMM and SOCAMM2 should not be framed as a simple winner-takes-all performance contest. MRDIMM prioritizes absolute CPU memory bandwidth, capacity and compatibility with the mature DIMM ecosystem. SOCAMM2 packages LPDDR5X in a compact, detachable module and prioritizes bandwidth per watt and physical density—an attractive objective for NVIDIA's vertically integrated, GPU-centric rack design. At the CPU level, a 16-channel Gen2 MRDIMM configuration delivers roughly 1,638 GB/s versus c.1,229 GB/s for eight SOCAMM2 modules on NVIDIA Vera. The capacity gap is larger: high-density MRDIMM configurations can exceed 16TB per CPU, compared with only 0.75TB SOCAMM2 used by Vera. SOCAMM2's clearest advantage is power efficiency. NVIDIA states that LPDDR5x delivers roughly 2x the bandwidth at the half the power of conventional DDR5 as baseline, implying 4x bandwidth per watt. The size of a SOCAMM2 module is roughly 30% of that of an RDIMM module. The smaller package can free board and airflow space.

In shorthand, SOCAMM2 optimizes GB/s per watt and physical compactness; MRDIMM optimizes total GB/s and GB per socket. But migration from DIMM to SOCAMM2 requires more than substituting one module for another.

EXHIBIT 18: configuration comparison between SOCAMM2 and DDR5 RDIMM



- ~133mm x ~31mm x ~2.5mm
- 288 pins on edge connector
- Vertical, insertion socket



- ~86mm x ~14mm x ~1mm
- 694 pins on back
- Horizontal, screw mounted

Source: AMD website, Bernstein analysis

EXHIBIT 19: MRDIMM maximizes absolute bandwidth and capacity; SOCAMM2 optimizes power efficiency and compactness

DERIVED METRIC	Aggregated bandwidth	÷	Power	=	Bandwidth per W	Superior	Moderate	Inferior
	DDR5 RDIMM (Gen 3)		DDR5 MRDIMM (Gen 2)		LPDDR5x SOCAMM2 for NV Vera (original design)		LPDDR5x SOCAMM2 for NV Vera (adjusted design)	
Data rate	6,400 MT/s (Gen 3)		12,800 MT/s (Gen 2)		9,600 MT/s			
Aggregated bandwidth for CPU	819.2 GB/s for 16-channel CPU (6400*64 bit*16-ch/8)		1,638.4 GB/s for 16-channel CPU (12800*64bit*16-ch/8)		1,228.8 GB/s for NV Vera (9600*128bit*8 modules/8)			
Memory capacity per module	16/32/64/96/128/256 GB		128/256/512 GB		192 GB		96 GB	
Total capacity per CPU	Up to 8TB+ (16-ch * 2 DIMM/Ch * 256GB)		Up to 16TB+ (16-CH * 2 DIMM/Ch * 512GB)		1.5TB (8 SOCAMM2 * 192GB)		0.75TB (8 SOCAMM2 * 96GB)	
Power efficiency	Moderate		Slightly higher than RDIMM, due to additional interface chips		Best-in-class: deliver 2x bandwidth at half the power of traditional DDR			
Bandwidth per W	0.25x (GB/s/W)		0.24x (GB/s/W)		Assuming in x (GB/s/W)			
Compactness	288 pins, requiring vertical slot space		Similar to RDIMM		Highly compact low-profile compression module, smaller footprint			
Price per GB	\$24.3		\$27.7 (small sampling price)		\$29 (small sampling price)			
Per module price	\$3,200+ (128GB)		\$3,550 (128GB)		\$5,600 (196GB)		\$2,800 (96GB)	

Source: NVIDIA reports, Bernstein analysis and estimates

HIGH SWITCHING COST AS THE MOAT OF DIMM ECOSYSTEM

The relevant switching cost sits at the CPU-platform level. A vendor must redesign the memory controller and PHY, package and I/O allocation, board layout, connector, power delivery, firmware, cooling, validation and service model. The redesign requires requalifying the supply chain and forfeiting backward compatibility. NVIDIA can justify this greenfield investment because it controls the GPU, CPU, interconnect, power and rack architecture. Intel, AMD and other ARM vendors already operate inside a mature DDR ecosystem and can obtain higher bandwidth through MRDIMM without absorbing the same system-level disruption.

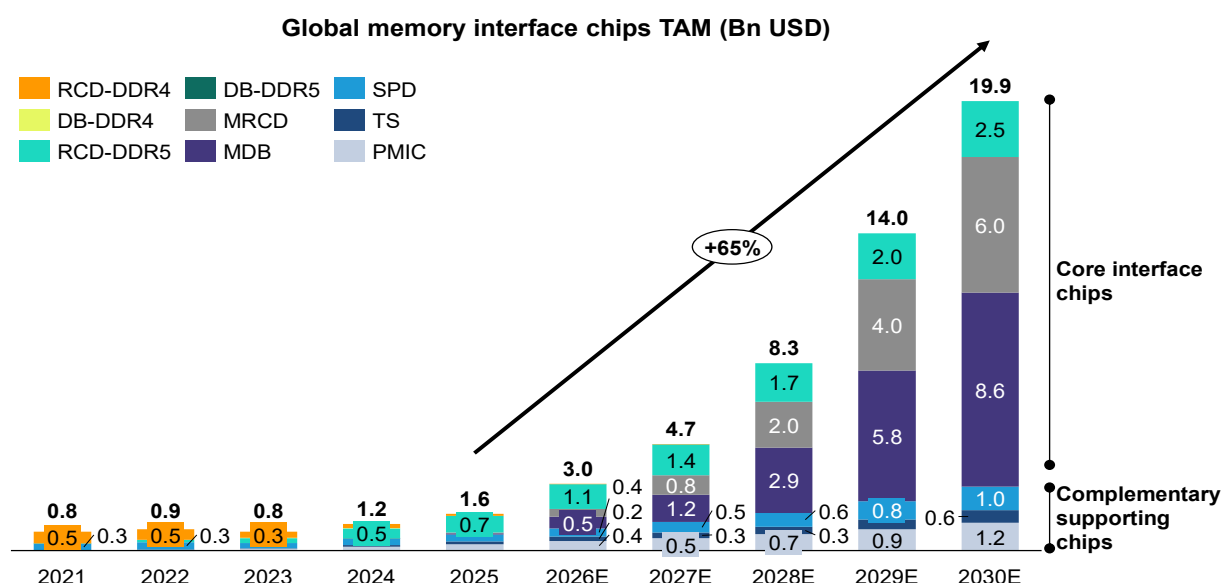
SOCAMM2 therefore remains a credible NVIDIA-specific solution rather than a broad substitute for MRDIMM. It also adds modest interface-chip content, but only a few dollars per module versus roughly USD70–80 for MRDIMM. We expect its impact on the global memory-interface-chip TAM to remain limited as long as deployment stays concentrated in NVIDIA's proprietary CPU platforms.

TAM OF MEMORY INTERFACE CHIPS FOR MRDIMM

We project the global DDR module interface chipset TAM to reach USD 20Bn in 2030, growing at a 65% CAGR from 2025. MRDIMM-related MRCD and MDB products contribute roughly 73% of the 2030 total, plus MSD contributed by supporting chips required by MRDIMM modules. This concentration reflects the unusually high interface-chip value per module and makes MRDIMM penetration the most sensitive variable in the memory interface chip TAM model.

Our projection excludes memory-interface components used in NVIDIA proprietary SOCAMM2, including voltage regulators and PMICs. We expect this exclusion to have only a LSD impact, because NVIDIA CPUs represent a small portion of global server CPU shipments and these components carry relatively low value compared to chipset used by DIMM. The core forecast therefore remains tied primarily to x86 and non-NVIDIA ARM platforms using DIMM-format DDR.

EXHIBIT 20: We raise our global memory interface TAM projection to USD 20Bn in 2030, with MRDIMM's MRCD and MDB contributing 73%, driven by significantly higher chipset value per module

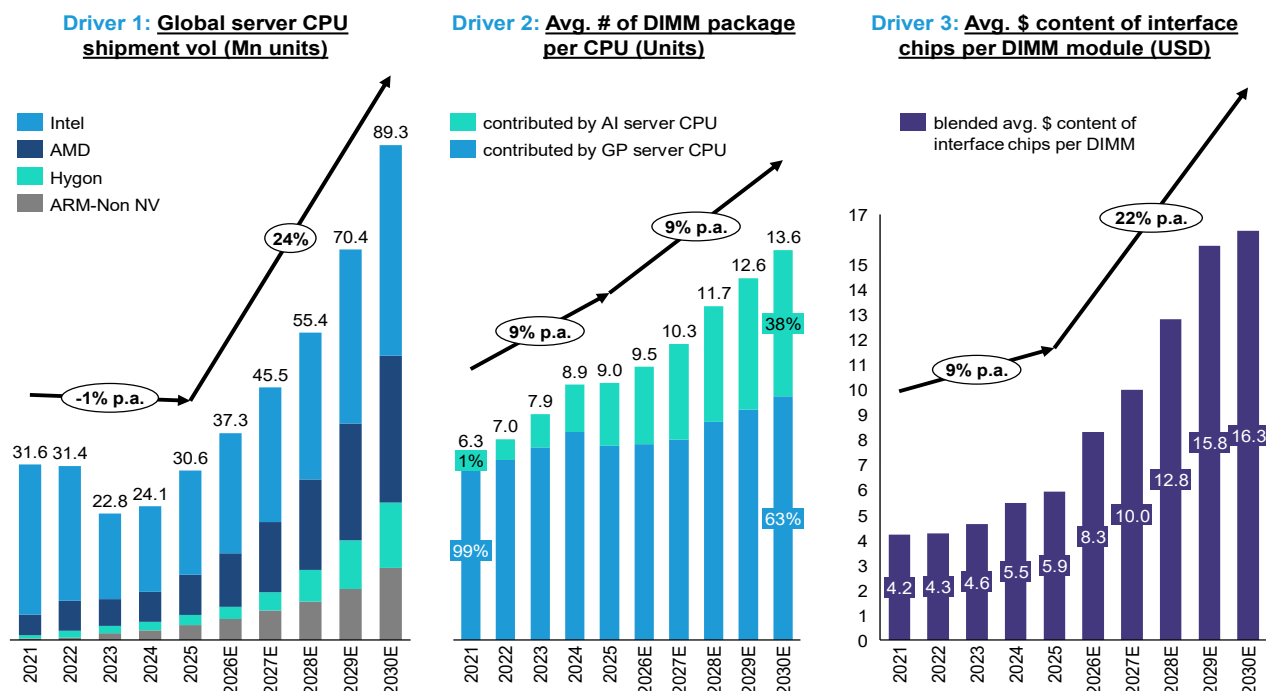


Our TAM forecast does not include memory interface chips used in NVIDIA proprietary CPUs, such as voltage regulators and PMICs. We expect the impact to be only LSD, as NVIDIA CPUs account for a small portion of server CPU shipments and the related components are of relatively low value.

Source: Mercury, TrendForce, Gartner, Bernstein analysis and estimates

Three multiplicative drivers underpinning memory-interface chip TAM will accelerate through the second half of the decade. Our TAM projection is the product of server CPU shipments, DIMM modules per CPU, and interface-chip value per DIMM. We project global server CPU shipments (excluding NVIDIA proprietary CPUs) to rise from 30.6Mn units in 2025 to 89.3Mn units in 2030, at a 24% CAGR. We forecast the average number of DIMM packages per CPU to rise from 9.5 to 13.6 in 2030, as 12- and 16-channel architectures gain share and AI systems run higher slot of occupancy. The third driver provides the largest amplification. We estimate blended interface-chip dollar content per DIMM to rise from USD 5.9 to USD 16.3 in 2030, as DDR5 generation upgrades and MRDIMM gains share. These factors are multiplicative rather than additive: more CPU sockets each carry more modules, and a larger portion of those modules carries substantially more silicon. This explains why the memory-interface-chip TAM grows faster than CPU units alone.

EXHIBIT 21: The expansion of the TAM is driven by three core factors, each of which is expected to accelerate in the second half of this decade relative to the past 5 years



Source: Gartner, Mercury, Bernstein analysis and estimates

EXHIBIT 22: The third driver provides the largest amplification: MRDIMM boosts memory interface chip dollar content by ~10x vs. RDIMM

Unit of chips	DDR4 RDIMM	DDR4 LRDIMM	DDR5 RDIMM	DDR5 MRDIMM
RCD	1	1	1	
DB		9		
MRCD				1
MDB				10
SPD	0 or 1	0 or 1	1	1
TS			2	2
PMIC			1	1
ASP of the whole memory interface chipset	USD 2-3	USD 12-14	USD 6-7	USD 70-80

This chart shows our estimation of the chip price in 2026, we expect the price to further drop in the next few years.

Source: Montage reports, Bernstein analysis

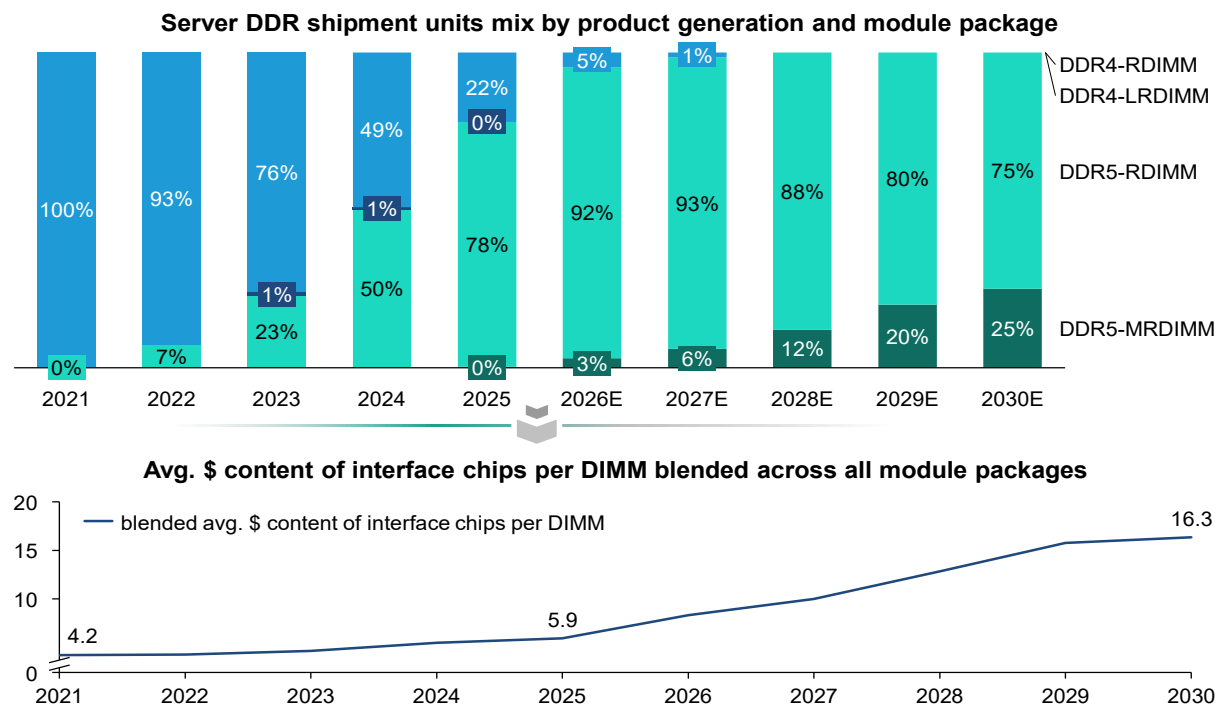
HISTORICAL DDR TRANSITIONS OFFER A REFERENCE

DDR4-to-DDR5 adoption shows how quickly a platform-mandated generation can move from negligible to dominant share. We forecast MRDIMM to reach 25%+ of server DDR DIMM units in 2030, still far below DDR5 RDIMM's c.80% share five years after initial ramp. MRDIMM is an optional performance tier and can coexist with RDIMM for workloads that do

not need the additional bandwidth.

MRDIMM also differs structurally from DDR4 LRDIMM, which never exceeded roughly 1% of server DIMM units. LRDIMM increased capacity but did not increase per-module bandwidth and added latency at a time when RDIMM was generally sufficient. MRDIMM addresses the binding AI-era constraint by roughly doubling per-module bandwidth, both Intel and AMD support the architecture, and hyperscale customers now run workloads that can convert bandwidth into throughput. These differences reduce the risk that MRDIMM remains a niche tier.

EXHIBIT 23: We expect MRDIMM penetration to surpass 25% by 2030, a level that remains modest compared with the adoption trajectory of DDR5 RDIMMs, whose unit share rose from neglectable to ~80% in just 5 years



Source: TrendForce, Bernstein analysis and estimates

SENSITIVITY CHECK

Our sensitivity analysis yields a 2030 TAM range of USD8–38bn across 60–120mn server CPUs and 10–40% MRDIMM penetration. The lower-left case is materially below our USD20bn base case, but still returns to approximately the prior-version TAM estimate.

We would monitor three signposts: OEM and CSP qualification beyond sample programs, module premium as supply scales, and the pace at which MRDIMM-capable Intel and AMD platforms enter high-bandwidth deployments. The thesis does not require MRDIMM to displace RDIMM universally. It requires the product to become common within a sufficiently large pool of host, recommendation, retrieval, CPU-inference and large-memory workloads—where a modest module premium can protect much more valuable system compute.

EXHIBIT 24: 2030 TAM Sensitivity Analysis: Even in a bear-case scenario, the global memory interface chip TAM reaches our prior-version projection, reinforcing a well-protected downside for Montage at current share price

2030 TAM sensitivity analysis (Bn USD)		Server CPU shipment in 2030 (excl. NV, Mn units)						
		60	70	80	89	100	110	120
MRDIMM penetration in 2030	10%	8	9	10	12	13	14	16
	15%	10	11	13	14	16	18	19
	20%	12	13	15	17	19	21	23
	25%	13	16	18	20	22	24	27
	30%	15	18	20	23	25	28	30
	35%	17	20	23	25	28	31	34
	40%	19	22	25	28	32	35	38

Source: Bernstein analysis and estimates

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 25: Montage income statement

INCOME STATEMENT	FY 2021	FY 2022	FY 2023	FY 2024	FY 2025	FY 2026	FY 2027	FY 2028	FY 2029	FY 2030
Revenue	2,562	3,672	2,286	3,639	5,456	7,324	13,662	25,322	42,075	58,408
Revenue YoY%		43%	(38%)	59%	50%	34%	87%	85%	66%	39%
Revenue QoQ%										
Quarterly revenue mix										
Cost of sales (enter as -)	(1,330)	(1,967)	(939)	(1,524)	(2,061)	(2,440)	(4,666)	(8,936)	(15,531)	(22,569)
Gross Profit	1,232	1,706	1,347	2,115	3,395	4,884	8,996	16,386	26,544	35,839
Gross profit margin%	48.1%	46.4%	58.91%	58.13%	62.2%	66.7%	65.8%	64.7%	63.1%	61.4%
YoY Bp Δ		-164 bps	1247 bps	-78 bps	410 bps	446 bps	-84 bps	-114 bps	-162 bps	-173 bps
Other Operating Income	76	1	2	12	116	43	71	71	71	71
Selling Expenses	(45)	(86)	(90)	(96)	(118)	(137)	(160)	(181)	(199)	(216)
General & admin expenses	(198)	(208)	(179)	(203)	(480)	(470)	(531)	(584)	(637)	(688)
Total SG&A	(243)	(295)	(269)	(299)	(598)	(606)	(691)	(765)	(836)	(904)
Research and development expenses	(370)	(563)	(682)	(763)	(915)	(1,139)	(1,378)	(1,626)	(1,919)	(2,264)
Other Operating Expense	(42)	(356)	(199)	(19)	(28)	(65)	(141)	(201)	(288)	(372)
Total operating expenses	(655)	(1,214)	(1,150)	(1,081)	(1,541)	(1,810)	(2,210)	(2,592)	(3,042)	(3,539)
Expenses analysis & assumptions										
Selling Expense YoY growth		92%	4%	7%	23%	16%	17%	13%	10%	8%
Admin Expense YoY growth		5%	(14%)	13%	137%	(2%)	13%	10%	9%	8%
SG&A Ratio	(9%)	(8%)	(12%)	(8%)	(11%)	(8%)	(5%)	(3%)	(2%)	(2%)
R&D YoY growth		52%	21%	12%	20%	24%	21%	18%	18%	18%
Financial income/ (expenses) as % of b		2%	3%	4%	4%	6%	21%	4%	4%	5%
Investment income as % of beginning b		(1%)	(0%)	(1%)	(1%)	(1%)	(4%)	(4%)	(4%)	(4%)
Other gain / (loss) as of sales	1%	(10%)	(9%)	(0%)	2%	(0%)	(1%)	(1%)	(1%)	(1%)
Total Operating Expenses Ratio	(26%)	(33%)	(50%)	(30%)	(28%)	(25%)	(16%)	(10%)	(7%)	(6%)
Operating profit	653	493	199	1,046	1,971	3,118	6,857	13,865	23,573	32,371
Operating profit margin	25.5%	13.4%	8.7%	28.8%	36.1%	42.6%	50.2%	54.8%	56.0%	55.4%
Interest Income (Expense)	72	81	179	230	269	512	700	861	1,153	1,587
Income (Loss) from Affiliates	(6)	(4)	(4)	(7)	(15)	(13)	(26)	(26)	(26)	(26)
Non-operating activities	196	843	98	143	95	491	--	--	--	--
Pretax profit	915	1,414	472	1,413	2,321	4,107	7,530	14,699	24,699	33,931
Taxes (enter expense as -)	(86)	(114)	(21)	(72)	(191)	(353)	(904)	(1,911)	(3,458)	(5,090)
Effective tax rate	(9.4%)	(8.1%)	(4.5%)	(5.1%)	(8.2%)	(8.6%)	(12.0%)	(13.0%)	(14.0%)	(15.0%)
Net Profit (incl. minority interest)	829	1,299	451	1,341	2,130	3,753	6,626	12,788	21,241	28,842
Loss/(gain) attributable to the noncontrolli	--	(0)	0	(71)	(106)	(109)	(106)	(106)	(106)	(106)
Minority interest as % of net profit	-	(0.0%)	0.1%	(5.3%)	(5.0%)	(2.9%)	(1.6%)	(0.8%)	(0.5%)	(0.4%)
Profit attributable to equity owners of	829	1,299	451	1,412	2,236	3,863	6,732	12,894	21,347	28,948
Net profit margin%	32.4%	35.4%	19.7%	38.8%	41.0%	52.7%	49.3%	50.9%	50.7%	49.6%
Net Profit YoY		57%	(65%)	213%	58%	73%	74%	92%	66%	36%
Extraordinary gain/ (loss)	--	--	--	--	--	--	--	--	--	--
Basic EPS	0.73	1.15	0.40	1.25	1.97	3.24	5.59	10.60	17.54	23.54
Diluted EPS	0.73	1.15	0.40	1.25	1.96	3.18	5.47	10.49	17.35	23.51

Source: Bloomberg, Bernstein analysis and estimate

EXHIBIT 26: **Montage balance sheet**

BALANCE SHEET	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Assets										
Cash & equivalents	5,087	5,834	5,666	6,699	8,471	16,419	19,557	25,782	35,173	48,371
Financial assets held for trading	414	1,610	1,703	1,783	823	2,917	2,917	2,917	2,917	2,917
Accounts & other receivables, prepaym	172	322	294	388	568	679	1,273	2,367	3,939	5,472
Inventories	375	738	482	352	896	910	2,329	4,500	7,854	11,432
Other current assets	1,375	132	150	239	276	531	638	752	887	1,040
Total current assets	7,423	8,636	8,296	9,461	11,034	21,456	26,714	36,318	50,770	69,231
Total PP&E, intangible assets, ROU an	1,178	779	1,094	1,445	1,454	1,692	1,480	1,364	1,311	1,301
Long-term equity investment	290	1,149	1,207	1,226	1,080	660	660	660	660	660
Goodwill	--	--	--	--	--	--	--	--	--	--
Other non-current assets	68	122	101	86	180	306	306	306	306	306
Total non-current assets	1,536	2,050	2,401	2,758	2,714	2,659	2,447	2,330	2,277	2,268
Total assets	8,959	10,686	10,698	12,219	13,748	24,114	29,161	38,648	53,047	71,499
Liabilities and Shareholder's Equity										
Accounts payables and other payables	424	605	376	642	755	1,346	2,618	5,057	8,827	12,849
Short-term borrowings	13	12	13	16	15	21	21	21	21	21
Contract Liabilities	--	--	--	--	1	3	3	3	3	3
Other current liabilities	--	23	2	22	0	0	0	0	1	1
Total Current Liabilities	437	639	391	680	770	1,370	2,642	5,082	8,852	12,874
Long-term borrowings	20	9	2	29	19	122	122	122	122	122
Other non-current liabilities	111	111	98	114	88	124	124	124	124	124
Total Non Current Liabilities	131	119	100	143	107	245	245	245	245	245
Total liabilities	568	759	491	822	877	1,615	2,887	5,327	9,097	13,119
Capital Reserve & Additional Add-in Ca	5,866	1,134	1,139	1,145	6,118	14,507	14,507	14,245	14,245	13,983
Less: Treasury Stock	(300)	(300)	(300)	(428)	(848)	(847)	(847)	(586)	(586)	(324)
Retained earnings / (accumulated defic	2,646	3,622	3,732	4,805	6,371	8,921	12,624	19,715	30,389	44,863
Other comprehensive income / (loss)	--	--	--	--	--	--	--	--	--	--
Other equity tools	179	5,456	5,621	5,881	1,283	51	229	290	352	413
Total Shareholders Equity	8,391	9,912	10,191	11,403	12,924	22,631	26,512	33,665	44,400	58,935
Minority shareholders' interest	--	15	15	(7)	(53)	(132)	(238)	(344)	(450)	(556)
Total Equity	8,391	9,927	10,207	11,397	12,871	22,499	26,274	33,321	43,950	58,380
Total Liabilities & Equity	8,959	10,686	10,698	12,219	13,748	24,114	29,161	38,648	53,047	71,499

Source: Bloomberg, Bernstein analysis and estimates

EXHIBIT 27: **Montage cash flow statement**

CASH FLOW STATEMENT	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Net Income	829	1,299	451	1,341	2,130	3,861	6,732	12,894	21,347	28,948
Depreciation and Amortization	59	52	105	141	120	180	655	587	551	538
SBC & Others	243	(271)	193	(23)	361	89	178	62	62	62
Change in net working capital	(155)	(392)	(18)	162	(668)	126	(849)	(939)	(1,291)	(1,241)
Cash from operating activities	975	689	731	1,691	2,048	4,256	6,716	12,603	20,669	28,306
Net capex	(153)	(226)	(424)	(379)	(266)	(418)	(444)	(470)	(498)	(528)
Cash paid for equity investment	--	--	--	--	--	--	--	--	--	--
Other investments	(72)	290	(150)	(64)	1,090	(1,379)	--	--	--	--
Cash from investing activities	(226)	63	(574)	(443)	824	(1,815)	(444)	(470)	(498)	(528)
Change in debt	--	(13)	(16)	(21)	--	113	--	--	--	--
Change in equity	(270)	82	(15)	83	(359)	7,110	--	(262)	--	(262)
Cash Paid for Distribution of Dividends, P	(339)	(339)	(341)	(338)	(670)	(1,740)	(3,029)	(5,802)	(10,674)	(14,474)
Other Cash Paid Related to Financing Ac	--	--	--	--	--	(52)	(106)	156	(106)	156
Cash from financing activities	(609)	(270)	(372)	(277)	(1,029)	6,332	(3,135)	(5,908)	(10,779)	(14,580)
Exchange Gains (losses) on Net Cash	(61)	268	47	62	(126)	(300)	--	--	--	--
Net change in cash during period	79	750	(168)	1,033	1,717	8,473	3,138	6,225	9,391	13,198

Source: Bloomberg, Bernstein analysis and estimates

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

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VALUATION METHODOLOGY**Montage Technology**

We set our A-share TP at CNY 400, based on 51x 2BF P/E. For the H shares, we set a target price of HKD 520, implying a 15% premium to our A-share TP (we use the CNY to HKD exchange rate at 1:1.13). The H share premium reflects that global investors favor Montage as a scarce China AI-exposed name without direct geopolitical risks, unlike many other Chinese semiconductor companies that face entity list restrictions or export control headwinds. At our target price, the implied P/E multiple on 2BF EPS for H shares is 59x.

RISKS**Montage Technology**

Key potential risks:

1. A decrease in memory and AIDC server demand;
2. Intensify competition that leads to share loss and lower margin;
3. Failure to introduce new products for the AIDC networking market.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

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Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

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- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

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Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

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Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

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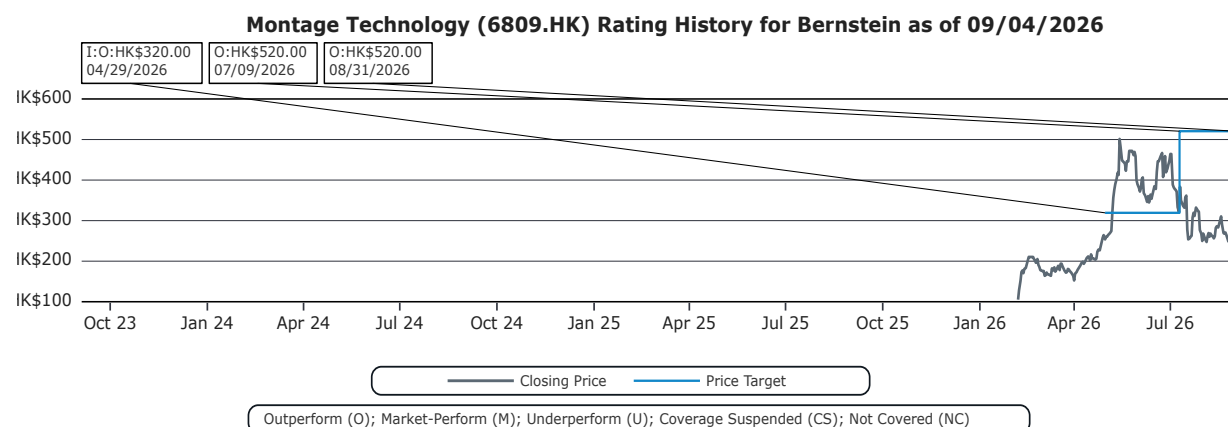
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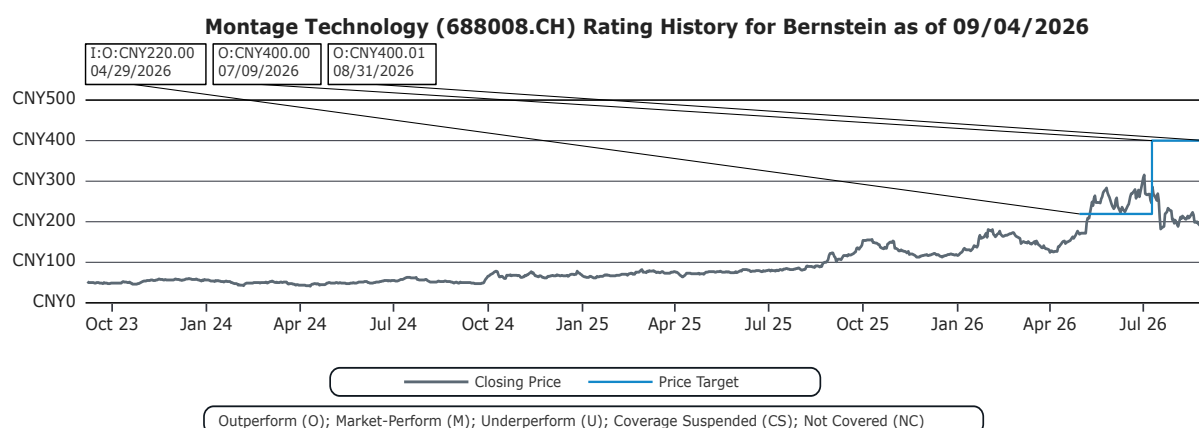
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Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
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